

Multiple Choice (10 marks)

1. The _____ can cost you money, by sending text messages from your mobile phone numbers.
 - (a) IM - Trojans
 - (b) Backdoor Trojans
 - (c) SMS Trojan
 - (d) Ransom Trojan
2. Can a stream cipher have perfect secrecy?
 - (a) Yes, if the PRG is really "secure"
 - (b) No, there are no ciphers with perfect secrecy
 - (c) Yes, every cipher has perfect secrecy
 - (d) No, since the key is shorter than the message
3. Which of the following regular expressions is equivalent to (describes the same set of strings as) $(a^* + b)^*(c + d)$?
 - (a) $a^*(c + d) + b(c + d)$
 - (b) $a^*(c + d)^* + b(c + d)^*$
 - (c) $a^*(c + d) + b^*(c + d)$
 - (d) $(a + b)^*c + (a + b)^*d$
4. What are the types of scanning?
 - (a) Port, network, and services
 - (b) Network, vulnerability, and port
 - (c) Passive, active, and interactive
 - (d) Server, client, and network
5. Which of the following is a remote Trojan?
 - (a) Troya
 - (b) DaCryptic
 - (c) BankerA
 - (d) Game-Troj

True / False (10 marks)

Answer True or False

6. True or False: Wireless traffic sniffing is a process used for analyzing wireless network data, aiding in forensic investigations and troubleshooting.
7. True or False: Freenet is a decentralized platform used for anonymous file sharing, functioning as part of the darknet.
8. True or False: In Message Confidentiality, the transmitted message must be understandable to both the sender and the translator.
9. True or False: Wireshark is a widely used tool for capturing and analyzing wireless network traffic.
10. True or False: Ruby and Python applications commonly encounter buffer overflow errors because of their dynamic typing features.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find all index positions of the minimum values in a given list.
12. Write a python function to find quotient of two numbers.

End of Paper